DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: X | **FULL MARKS: 40**
SUBJECT: CHEMISTRY | **DATE: 17.10.2022**

General Instructions:

- **The paper consists of three printed pages.**
- **Answers should be to the point.**
- **Question numbers should be copied carefully while answering the questions.**
- **Marks will be deducted for spelling errors.**

Question:1
Select the correct option: **[5]**

i) **Two consecutive members of a homologous series differ in their formula by:**
A) CH_2
B) CH_3
C) C_2H_2
D) C_3H_3

ii) **The difference in molecular mass between butyne and pentyne is:**
A) 13
B) 14
C) 12
D) 11

iii) **The metal which is not present in stainless steel is:**
A) Aluminium
B) Iron
C) Chromium
D) Manganese

iv) **The molecular mass of octane is 114, its vapour density is:**
A) 56
B) 57
C) 55
D) 54

v) **Which of the following is not a sulphide ore?**
A) Haematite
B) Galena
C) Cinnabar
D) Zinc blende

Question:2

Match the alloys given in column 1 with the uses given in column 2: **[5]**

Column 1		Column 2	
i)	Duralumin	A)	Electrical fuse
ii)	Solder	B)	Surgical instruments
iii)	Brass	C)	Aircraft body
iv)	Bronze	D)	Decorative articles
v)	Stainless steel	E)	Statues and medals

Question: 3

A gaseous hydrocarbon contains 82.76% of carbon, given that its vapour density is 29 find its molecular formula. [C = 12, H = 1] **[5]**

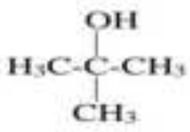
Question: 4

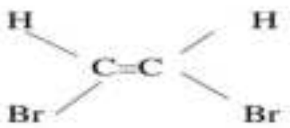
Write the structural formula of the following: **[5]**

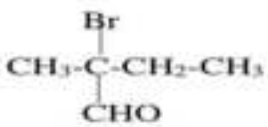
- **i) Methanal**
- **ii) Propan-2-ol**
- **iii) Acetone**
- **iv) Ethanoic acid**
- **v) 1, 2-Dichloroethane.**

Question: 5

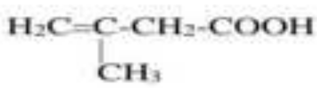
Write the IUPAC name of the following: **[5]**

1. 

2. 

3. 

4. $H_3C-C\equiv C-CH_3$

5. 

Question: 6

i) **1250 cc of oxygen was burnt with 300 cc of ethane. Calculate the volume of unused oxygen and the volume of carbon dioxide formed. The equation for the reaction is: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$** **[3]**

ii) **1000 ml of a gas X weighs 1 g, calculate the vapour density and molecular mass of the gas if 1 litre of hydrogen weighs 0.09 g at STP.** **[2]**

Question: 7

With respect to extraction of aluminium answer the following: **[10]**

i) **Name the chief ore of aluminium and also write its formula.**

ii) **Why are the anodes replaced periodically during electrolysis of fused alumina. Name one main impurity associated with the ore.**

iii) **Write three balanced equations for the reaction taking place during the concentration of the ore.**

iv) **Explain why aluminium cannot be obtained by reducing its oxide using reducing agents.**

v) **Mention the role of the following in Hall Heroult's process:**

   a) **Graphite**

   b) **Powdered coke**